RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

Name : Prajit.R

Year / Branch / Section : 2nd Year, Computer Science And Design

University Register No. : 2116251701040

College Roll No. : 251701040

Semester : 3rd Semester

Academic Year : 2026-2027

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 8:44 PM
State	Finished
Completed on	Saturday, 10 October 2026, 8:45 PM
Time taken	50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, i, j;
    scanf("%d", &n);

    int arr[n];

    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                printf("%d", arr[i]);
                return 0;
            }
        }
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 8:47 PM
State	Finished
Completed on	Saturday, 10 October 2026, 8:48 PM
Time taken	42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, i, j;
    scanf("%d", &n);

    int arr[n];

    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                printf("%d\n", arr[i]);
                return 0;
            }
        }
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 8:49 PM
State	Finished
Completed on	Saturday, 10 October 2026, 8:50 PM
Time taken	41 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2, i, j;
        scanf("%d", &n1);

        int *a = malloc(n1 * sizeof(int));

        for (i = 0; i < n1; i++) {
            scanf("%d", &a[i]);
        }

        scanf("%d", &n2);
        int *b = malloc(n2 * sizeof(int));

        for (i = 0; i < n2; i++) {
            scanf("%d", &b[i]);
        }

        i = 0;
        j = 0;
        int first = 1;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                if (i == 0 || a[i] != a[i - 1]) {
                    if (!first) {
                        printf(" ");
                    }
                    printf("%d", a[i]);
                    first = 0;
                }
                i++;
                j++;
            } else if (a[i] < b[j]) {
                i++;
            } else {
                j++;
            }
        }

        printf("\n");
        free(a);
        free(b);
    }

    return 0;
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 8:52 PM
State	Finished
Completed on	Saturday, 10 October 2026, 8:57 PM
Time taken	4 mins 41 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2, i, j, first = 1;
        int a[10000], b[10000];

        scanf("%d", &n1);
        for (i = 0; i < n1; i++)
            scanf("%d", &a[i]);

        scanf("%d", &n2);
        for (i = 0; i < n2; i++)
            scanf("%d", &b[i]);

        i = 0;
        j = 0;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                if (first || a[i] != a[i - 1]) {
                    if (!first)
                        printf(" ");
                    printf("%d", a[i]);
                    first = 0;
                }
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 8:59 PM
State	Finished
Completed on	Saturday, 10 October 2026, 9:00 PM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i, j;
    scanf("%d", &n);

    int *A = malloc(n * sizeof(int));

    for (i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    long long k;
    scanf("%lld", &k);

    i = 0;
    j = 1;

    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }

        long long diff = (long long)A[j] - A[i];

        if (diff == k) {
            printf("1\n");
            free(A);
            return 0;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }
    }

    printf("0\n");
    free(A);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 9:05 PM
State	Finished
Completed on	Saturday, 10 October 2026, 9:06 PM
Time taken	1 min 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i;
    scanf("%d", &n);

    int *A = malloc(n * sizeof(int));

    for (i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    long long k;
    scanf("%lld", &k);

    int left = 0, right = 1;

    while (left < n && right < n) {
        if (left == right) {
            right++;
            continue;
        }

        long long diff = (long long)A[right] - A[left];

        if (diff == k) {
            printf("1\n");
            free(A);
            return 0;
        } else if (diff < k) {
            right++;
        } else {
            left++;
        }
    }

    printf("0\n");
    free(A);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review